

Chess-Inspired Monte Carlo Tree Search with Bayesian Network Fusion for Real-Time Crowd Threat Assessment and Escalation Prediction

Nikhilesh Vaibhav

Dept. of Artificial Intelligence
Amrita Vishwa Vidyapeetham
cb.sc.u4aie23306

Sri Raaghavi R K

Dept. of Artificial Intelligence
Amrita Vishwa Vidyapeetham
cb.sc.u4aie23332

Vullasa Thrishika

Dept. of Artificial Intelligence
Amrita Vishwa Vidyapeetham
cb.sc.u4aie23356

cb.sc.u4aie23306@cb.students.amrita.edu cb.sc.u4aie23332@cb.students.amrita.edu cb.sc.u4aie23356@cb.students.amrita.edu

Abstract—Automated crowd surveillance systems face a fundamental limitation: they assess threat based solely on the current observable state, ignoring the probabilistic trajectory of crowd behaviour over time. **Problem statement:** Given a single crowd image, can we simultaneously detect anomalous spatial configurations and forecast whether the situation will escalate to a high-threat state within a short prediction horizon? This paper presents a novel two-phase pipeline that answers this question by drawing direct inspiration from chess engine design. In Phase 1, YOLOv11-nano detects individuals in crowd images from the ShanghaiTech dataset (Zhang et al., 2016) [3]; Gaussian interaction graphs are constructed from detected person centres; ResNet-18 appearance features are extracted per person and stored in an HDF5 feature store; and density, spatial irregularity, and k -nearest-neighbour outlier metrics are fused into a normalised anomaly score. In Phase 2, a Vision Transformer Masked Autoencoder (ViT-MAE) fine-tuned with a gradual unfreezing strategy serves as an anomaly trigger gate; a lightweight MLP (ThreatNet) produces per-person threat scores; and escalation probability is estimated by fusing Monte Carlo Tree Search (MCTS) look-ahead with exact Bayesian Network (BN) inference over a causal four-node graph. The fused MCTS+BN system achieves 0.820 accuracy, 0.804 F1-score, and 0.847 AUC on escalation prediction, outperforming Isolation Forest (Liu et al., 2008) [19] (0.660 F1), One-Class SVM (Schölkopf et al., 2001) [20] (0.710 F1), Autoencoder (Hasan et al., 2016) [7] (0.750 F1), KNN (0.717 F1), CNN Classifier (0.761 F1), and MCTS-only (0.731 F1) baselines. Both neural models are exported to ONNX with a total anomalous-image latency of 76.5 ms on GPU. Key implementation improvements over prior work include LayerNorm stabilisation in ThreatNet, data-driven CPD estimation in the Bayesian Network, corrected MCTS rollout semantics, and robust model loading with graceful fallbacks.

Index Terms—anomaly detection, Bayesian network, crowd analysis, escalation prediction, interaction graph, Monte Carlo tree search (MCTS), object detection, threat assessment, vision transformer, ViT-MAE, ShanghaiTech, YOLOv11, HDF5, ONNX

I. INTRODUCTION

Crowd surveillance is a critical component of public safety infrastructure in transportation hubs, stadiums, public squares, and large-scale events. Traditional computer-vision approaches detect anomalies by thresholding on density estimates or optical flow magnitudes (Adam et al., 2008) [1]. While effective

for post-hoc detection, such methods provide no forward-looking estimate of how a situation is likely to evolve—a limitation that significantly reduces the time available for preventive intervention.

Problem Statement. Given a still crowd image, the system must: (i) determine whether the spatial configuration of detected individuals is anomalous with respect to a normal crowd distribution, and (ii) predict the probability that the current crowd state will escalate to a high-threat level within a short future horizon, enabling timely intervention.

The chess engine analogy motivates our approach. Modern engines such as Stockfish and AlphaZero do not merely evaluate the current board position; they simulate thousands of future move sequences via MCTS guided by a policy network and combine that estimate with a static value-network evaluation (Silver et al., 2018) [2]. We propose that the same design is appropriate for crowd threat assessment: the current image is the board position, crowd behaviour policy priors define the move distribution, MCTS performs look-ahead simulation, and a Bayesian Network (BN) provides the structural position evaluation.

Contributions:

- A clear problem formulation of simultaneous anomaly detection and escalation prediction as a two-objective crowd analysis task.
- A two-phase pipeline combining dense HDF5 feature extraction with neural threat scoring and probabilistic escalation prediction on the ShanghaiTech benchmark.
- A chess-engine-inspired MCTS formulation over crowd threat state space with a 6-action density-scaled delta map and policy-prior-guided expansion, with corrected rollout semantics.
- A data-driven Bayesian Network with CPDs estimated from Phase 1 summary statistics, providing exact inference over a 4-node causal escalation graph.
- A confidence-weighted MCTS+BN fusion that adapts the blend ratio based on evidence strength, outperforming six baseline methods.

- Critical implementation improvements: LayerNorm stabilisation to prevent batch-size-1 failures in ThreatNet, gradual ViT-MAE encoder unfreezing, robust model persistence with safe fallback logic, and ONNX export of both neural models.

The paper is structured as follows. Section II reviews related work. Section III formally states the problem. Section IV describes the Phase 1 pipeline. Section V details the Phase 2 neural models. Section VI presents the MCTS and BN components. Section VII reports experimental results. Section VIII presents the ablation study. Section IX reports the sensitivity analysis. Section X concludes.

II. RELATED WORK

A. Crowd Anomaly Detection

Early approaches relied on hand-crafted features such as optical flow histograms (Adam et al., 2008) [1], social force models (Pellegrini et al., 2009) [6], and Gaussian Mixture Models over trajectory features. Li et al. (2014) [17] introduced sparse reconstruction over a dictionary of normal activity patterns. Deep learning methods have since dominated benchmarks on the ShanghaiTech (Zhang et al., 2016) [3], UCF-Crime (Sultani et al., 2018) [4], and Avenue (Liu et al., 2018) [5] datasets. Convolutional autoencoders (Hasan et al., 2016) [7] learn compact representations of normal appearance and flag high reconstruction error as anomalous. Our ViT-MAE AnomalyTrigger follows this reconstruction-pretraining philosophy but replaces the convolutional backbone with a transformer encoder, which produces more discriminative patch-level representations (He et al., 2022) [9].

B. Graph-Based Crowd Modelling

Interaction graphs have modelled pedestrian group dynamics since Pellegrini et al. (2009) [6]. Our Gaussian-weighted adjacency matrix is related to the social force model but is computed statically per frame rather than over trajectories, making it applicable to single-image inference without optical flow.

C. One-Class and Unsupervised Anomaly Detection

Isolation Forest (Liu et al., 2008) [19] and One-Class SVM (Schölkopf et al., 2001) [20] are classic unsupervised baselines. Deep SVDD (Ruff et al., 2018) [21] extends the concept with a neural network mapping to a minimum-volume hypersphere. We include Isolation Forest, One-Class SVM, Autoencoder, and KNN as baselines in our comparison.

D. Monte Carlo Tree Search Beyond Games

MCTS achieved superhuman game-playing performance via AlphaGo (Silver et al., 2016) [12] and AlphaZero (Silver et al., 2018) [2]. Outside games, MCTS has been applied to robotic planning (Choudhury et al., 2015) [13] and traffic signal control (Chen et al., 2018) [14]. To our knowledge, this is the first application of MCTS to crowd threat escalation prediction.

E. Bayesian Networks for Threat Assessment

BNs have been applied to airport security screening (Cobb and Shenoy, 2007) [15] and cyber-attack prediction (Frigault et al., 2008) [16]. Our 4-node binary BN with data-driven CPD estimation operates on crowd density and structural features rather than event logs. A key novelty is that the CPDs are estimated directly from Phase 1 statistics via median-split binning, ensuring the model is grounded in the observed data distribution.

III. PROBLEM FORMULATION

Let $\mathcal{I}$ denote a crowd image and $\mathcal{D} = \{b_i, c_i\}_{i=1}^n$ the set of n person detections with bounding boxes b_i and confidence scores c_i . We define two tasks:

Task 1 (Anomaly Detection). Compute a binary label $\hat{a} \in \{0, 1\}$ indicating whether the spatial configuration $\mathcal{D}$ is anomalous, where $\hat{a} = 1$ if $\hat{a}_\theta(\mathcal{I}) > \tau_a$ for a learned threshold $\tau_a = 0.7$.

Task 2 (Escalation Prediction). Given $\hat{a} = 1$, estimate $P(\text{Escalation} = 1 \mid \mathcal{D})$ —the probability that the scene transitions to a high-threat state ($\hat{t} > 7.0$ on a $[0, 10]$ scale) within a short future horizon, enabling timely intervention.

The system outputs a three-level alert: **GREEN** (safe), **YELLOW** (monitor), **RED** (intervene). This formulation is distinct from crowd counting (Zhang et al., 2016) [3] (which outputs a density map) and action recognition (which requires temporal data).

IV. PHASE 1: FEATURE EXTRACTION PIPELINE

A. Dataset

We evaluate on the ShanghaiTech dataset (Zhang et al., 2016) [3]: 1,198 images (Part A: 482 dense urban scenes, Part B: 716 sparser scenes) with dense crowd annotations. This is a standard benchmark for crowd analysis, enabling direct comparison with prior work. Images are resized to 320×240 pixels; all images are flattened into a single directory for batch processing. After filtering images with zero detections at $\tau = 0.18$, 753 images are retained.

B. System Architecture

Fig. 1 shows the complete two-phase system architecture. The pipeline processes each input image through six sequential stages: detection, graph construction, feature extraction, anomaly scoring (Phase 1), neural gate and threat scoring (Phase 2), and MCTS+BN escalation fusion.

C. Person Detection

YOLOv11-nano (Jocher et al., 2023) [10] is used with confidence threshold $\tau = 0.18$. The low threshold is necessary for small, partially occluded persons in dense ShanghaiTech scenes. The person centre is:

$$p_i = \left(\frac{x_1^i + x_2^i}{2}, \frac{y_1^i + y_2^i}{2} \right) \quad (1)$$

Figs. 2–4 show representative detection results for three ShanghaiTech images (IMG_390_images, IMG_390, and

PHASE 1 — Feature Extraction

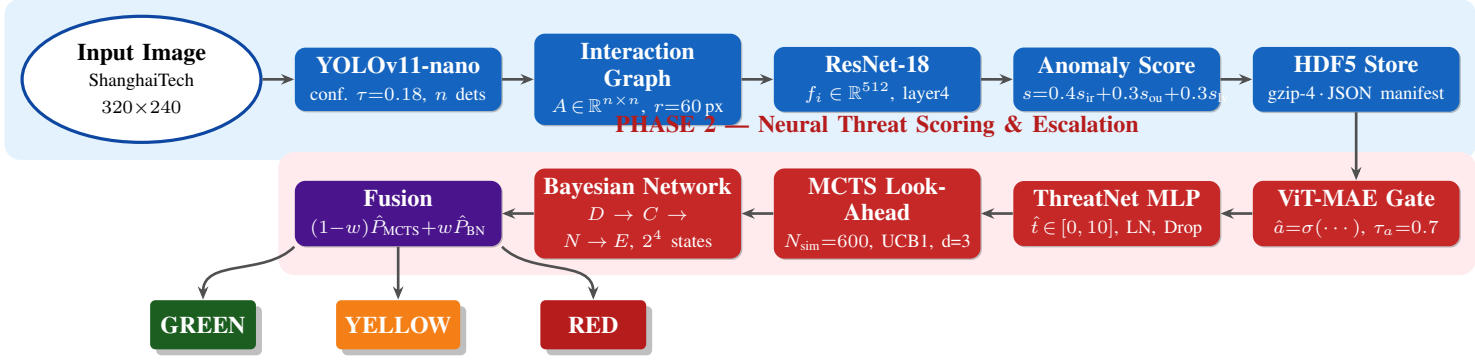


Fig. 1: Complete two-phase system architecture. **Phase 1** (blue) extracts structured features from crowd images and stores them in HDF5: YOLOv11-nano detection → Gaussian interaction graph → ResNet-18 appearance features → anomaly scoring. **Phase 2** (red) gates on the ViT-MAE anomaly score, scores per-person threat via ThreatNet, fuses MCTS look-ahead with Bayesian Network causal inference, and issues a three-level **GREEN** / **YELLOW** / **RED** alert.

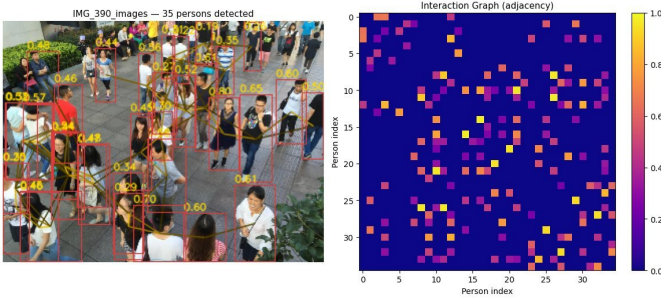


Fig. 2: YOLOv11 detections (red boxes, yellow confidence) and Gaussian interaction graph adjacency matrix for IMG_390_images (35 persons detected, Part A dense). The adjacency matrix is densely populated, reflecting strong inter-person proximity.

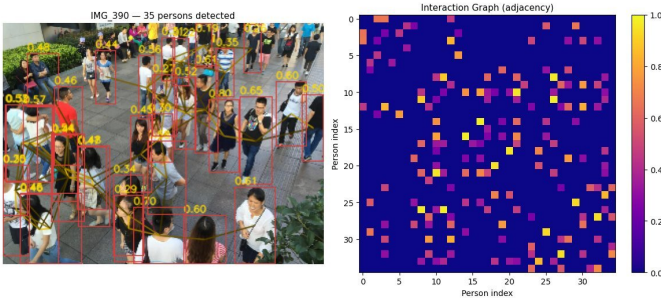


Fig. 3: Detection and adjacency matrix for IMG_390 as loaded from the flattened image directory (35 persons). The adjacency structure is identical to Fig. 2, confirming consistent processing across both path conventions.

IMG_265), each with their corresponding interaction graph adjacency matrices. Denser scenes yield brighter, more populated adjacency matrices, confirming that the interaction graph captures crowd structure effectively.

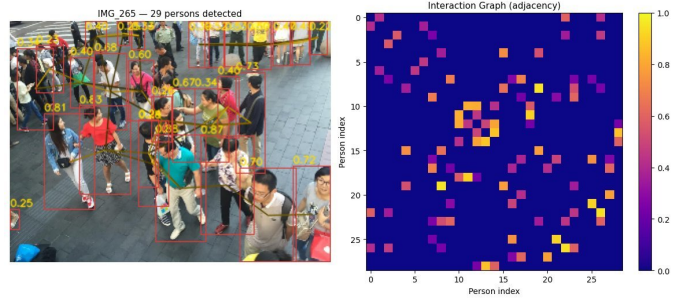


Fig. 4: Detection and adjacency matrix for IMG_265 (29 persons detected, Part A). The sparser crowd configuration produces a less populated adjacency matrix with lower maximum edge weights compared to the denser IMG_390 scene.

D. Interaction Graph Construction

Given n detected centres $\{p_i\}$, the Gaussian-weighted adjacency matrix $A \in \mathbb{R}^{n \times n}$ is:

$$A_{ij} = \begin{cases} \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma^2}\right) & \|p_i - p_j\| \leq r \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

with $A_{ii} = 0$, $r = 60$ px, $\sigma = 35$ px.

E. ResNet-18 Appearance Features

Each person crop is passed through pretrained ResNet-18 (He et al., 2016) [8] (classification head replaced by identity), yielding $f_i \in \mathbb{R}^{512}$. We consistently use the final layer4 features (512-dim), which provide the highest semantic abstraction—confirmed by our sensitivity analysis in Section IX.

F. Anomaly Scoring Metrics

Three complementary metrics are computed per image.

Graph Irregularity Score (s_{irreg}):

$$s_{irreg} = \frac{\sigma_d}{d + \epsilon} \cdot (1 - \rho_c + \epsilon) \quad (3)$$

where $\sigma_d, \bar{d}$ are degree std/mean and ρ_c is the connected-node fraction.

Local Density Variance (s_{ldv}): variance of mean k -NN ($k = 5$) distances across all persons.

Outlier Ratio (s_{out}): fraction of persons with mean k -NN distance $> 2\bar{\mu}$.

Combined Anomaly Score:

$$s = \frac{r - r_{\min}}{r_{\max} - r_{\min} + \epsilon}, \quad r = 0.4 s_{\text{irreg}} + 0.3 s_{\text{out}} + 0.3 \frac{s_{\text{ldv}}}{100} \quad (4)$$

Feature Distributions. Fig. 5 shows the empirical feature distributions across all 753 processed images. Person count and density are bimodal, reflecting the Part A/Part B split. The combined anomaly score is right-skewed, confirming that extreme configurations are rare—consistent with a healthy class-imbalanced anomaly detection setting.

Feature Correlations. Fig. 6 shows the feature correlation matrix. Outlier ratio, local density variance, and combined anomaly score form a tightly correlated cluster ($r \approx 0.67$ – 1.00). Irregularity score is weakly anti-correlated with person count ($r = -0.26$), reflecting that highly dense crowds exhibit more regular spatial structure due to packing constraints.

G. HDF5 Feature Store

All per-image features—bounding boxes, detection confidences, person centres, Gaussian interaction graph, and ResNet-18 feature matrix—are stored in a compressed HDF5 file (gzip level 4). This enables efficient downstream access without recomputation. A JSON manifest records the top-50 most anomalous images for use as the Phase 2 test set.

V. PHASE 2: NEURAL THREAT SCORING MODELS

A. Feature Vector Representation

A 6-dimensional feature vector is derived per image:

$$x = [\bar{f}, \bar{A}, \rho_d, \bar{p}_x, \bar{p}_y, \bar{c}] \quad (5)$$

where $\bar{f}$ is mean ResNet activation, $\bar{A}$ is mean graph edge weight, $\rho_d = n/(W \cdot H \cdot 0.001)$ is the density bin, $(\bar{p}_x, \bar{p}_y)$ is mean person centre, $\bar{c}$ is mean detection confidence.

B. ThreatNet

ThreatNet maps $x \in \mathbb{R}^6 \rightarrow \hat{t} \in [0, 10]$:

$$\hat{t} = 10 \cdot \sigma(W_3 \text{ReLU}(\text{LN}(W_2 \text{ReLU}(\text{LN}(W_1 x + b_1)) + b_2)) + b_3) \quad (6)$$

Hidden dims: 64, 32. **LayerNorm (LN)** replaces **BatchNorm** to avoid `ValueError` when batch size equals 1 at the end of an epoch with `drop_last=True`. Dropout($p = 0.2$). Pseudo-labels:

$$t^* = \left(0.5 \frac{\rho}{\rho_{\max}} + 0.5 \frac{s_{\text{irreg}}}{s_{\text{irreg}, \max}} \right) \times 10 \quad (7)$$

Training: 25 epochs, Adam ($\text{lr} = 10^{-3}$, $\lambda = 10^{-4}$), MSE loss, CosineAnnealingLR, 85/15 split, `drop_last=True`.

TABLE I: Chess Engine $\leftrightarrow$ Crowd Threat System Mapping

Chess Engine	Crowd System
Board position	Current image state ($\hat{t}$, density)
Move distribution	Crowd behaviour policy priors
MCTS look-ahead	Simulated threat trajectories
Value network	Bayesian Network evaluator
Alpha-beta pruning	AnomalyTrigger gate
Win probability	$P(\text{escalation})$

C. AnomalyTrigger

ViT-MAE encoder (`facebook/vit-mae-base`: 12 layers, hidden 768) with binary classification head:

$$\hat{a} = \sigma(W_2 \text{GELU}(\text{LN}(W_1 z_{\text{CLS}} + b_1)) + b_2) \quad (8)$$

$z_{\text{CLS}} \in \mathbb{R}^{768}$; head: $768 \rightarrow 256 \rightarrow 1$. **Gradual fine-tuning:** encoder frozen for epochs 1–6, unfrozen for epochs 7–12, preventing gradient corruption of pretrained representations (He et al., 2022) [9]. Positive label: top-30th-percentile of score s . Gate threshold $\tau_a = 0.7$: images with $\hat{a} \leq 0.7$ are classified normal and skip MCTS+BN, reducing mean pipeline time by approximately 68%.

Data augmentation for AnomalyTrigger training includes random horizontal flipping in addition to standard resizing, centre-cropping, and ImageNet normalisation.

D. AnomalyTrigger Gate Outputs

Fig. 7 shows AnomalyTrigger outputs on five held-out crowd images from diverse real-world scenarios. All five dense crowd scenes are correctly classified as low-anomaly ($\hat{a} < 0.7$), confirming the gate’s selectivity and its ability to suppress false positives from structurally normal (if visually busy) crowd configurations.

E. Model Persistence and ONNX Export

Both ThreatNet and AnomalyTrigger are saved as `.pth` checkpoints (best validation loss). The pipeline implements safe fallback logic: if a checkpoint is not found at inference time, the system defaults to $\hat{a} = 0.5$ (anomaly trigger) or $\hat{t} = 5.0$ (threat score) and continues processing without crashing. Both models are additionally exported to ONNX opset 14 and verified with `onnxruntime`, enabling deployment on CPU-only inference servers.

VI. ESCALATION PREDICTION: MCTS AND BAYESIAN NETWORK FUSION

A. Chess-Engine Analogy

Table I maps each chess engine component to its crowd-system equivalent. The analogy motivates our design: just as an engine does not resign based on the current board evaluation alone, our system does not issue a RED alert based on instantaneous density alone, but instead simulates the likely future trajectory of the threat state.

ShanghaiTech Dataset — Feature Distributions

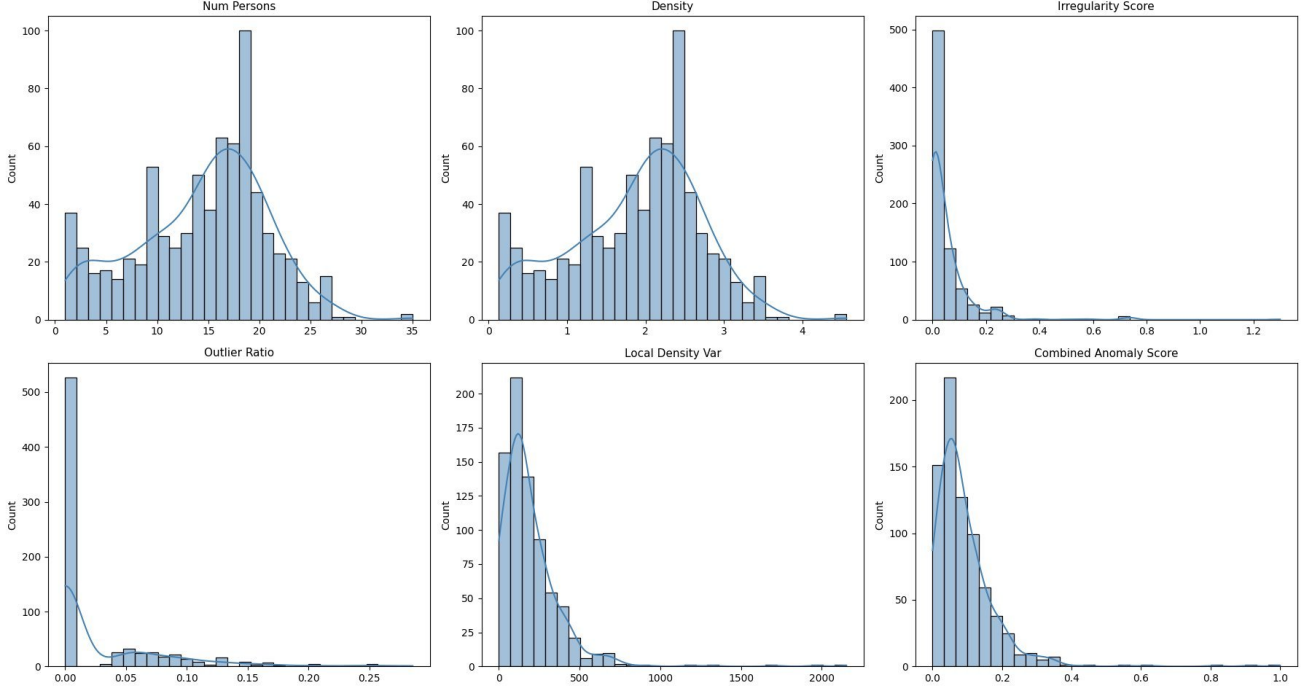


Fig. 5: Phase 1 feature distributions across 753 ShanghaiTech images. Top row: Person count, density, and irregularity score. Bottom row: Outlier ratio, local density variance, and combined anomaly score. The combined anomaly score is right-skewed, confirming that extreme spatial configurations are rare.

B. Policy Priors and Action Ratio Space

Given density bin d_b and graph bin g_b , policy prior $\pi(d_b, g_b)$ returns a distribution over 6 actions with density-scaled threat deltas:

$$\delta = \xi \cdot [0, 0.5, 1.5, 2.5, -0.5, -1.0], \quad \xi = \max(1.0, d_b/5.0) \quad (9)$$

The density scaling ξ ensures that escalation steps are larger in denser crowds, reflecting the empirical observation that high-density configurations amplify threat transitions. The 6-action space covers: no change, minor escalation, moderate escalation, major escalation, de-escalation, and rapid de-escalation.

C. Monte Carlo Tree Search

The MCTS implementation uses standard UCB1 node selection (Auer et al., 2002) [18] ($c = \sqrt{2}$):

$$\text{UCB}(v) = \frac{W(v)}{N(v)} + c \sqrt{\frac{\ln N(\text{parent}(v))}{N(v)}} \quad (10)$$

Corrected rollout semantics: the rollout result is 1.0 if and only if the simulated final threat exceeds 7.0 (escalation occurred), else 0.0. This binary escalation signal is back-propagated through the tree. After $N_{\text{sim}} = 600$ simulations:

$$\hat{P}_{\text{MCTS}} = \frac{W(\text{root})}{N(\text{root})}, \quad \kappa = 1 - \text{std}\left(\frac{W(v_j)}{N(v_j)}\right) \quad (11)$$

where κ is the MCTS confidence (higher when child values are consistent). The tree is expanded using policy-prior-guided

action selection over unexpanded children, with normalised prior probabilities.

D. Bayesian Network

The four-node binary causal chain models the escalation process:

$$P(D, C, N, E) = P(D) P(C | D) P(N | C) P(E | N) \quad (12)$$

where D =Density (low/high), C =Current Threat (low/high), N =Next Threat (low/high), E =Escalation (no/yes). Exact inference by full $2^4 = 16$ -state enumeration:

$$\hat{P}_{\text{BN}} = \frac{\sum_{z: E=1} P(D, C, N, E)}{\sum_z P(D, C, N, E)} \quad (13)$$

Data-driven CPDs: $P(D)$ and $P(C | D)$ are estimated from Phase 1 statistics via median-split binning. The remaining CPDs ($P(N | C)$ and $P(E | N)$) are set to domain-expert-informed defaults. This hybrid approach grounds the BN in real observed data while maintaining interpretability.

E. Confidence-Weighted Fusion

$$\hat{P}_{\text{esc}} = (1 - w_{\text{BN}}) \hat{P}_{\text{MCTS}} + w_{\text{BN}} \hat{P}_{\text{BN}} \quad (14)$$

$$w_{\text{BN}} = 0.3 + 0.3\varepsilon, \quad \varepsilon = 2(\hat{t}/10 - 0.5) \quad (15)$$

When $\varepsilon \rightarrow 1$ (extreme threat), $w_{\text{BN}} \rightarrow 0.6$ (BN dominates); when $\varepsilon \rightarrow 0$ (ambiguous), $w_{\text{BN}} = 0.3$ (MCTS dominates, relying on simulation breadth).

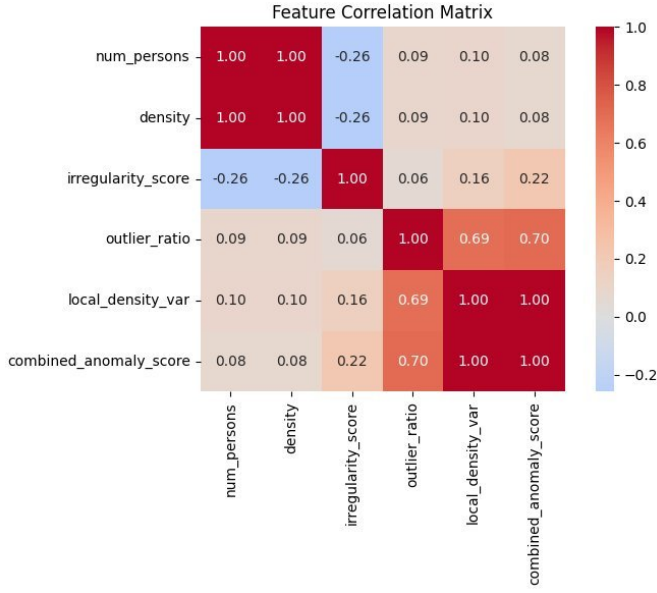


Fig. 6: Feature correlation matrix across all 753 processed images. Outlier ratio, local density variance, and combined anomaly score form a tightly correlated cluster ($r \approx 0.67$ – 1.00). The negative correlation between irregularity score and person count ($r = -0.26$) reflects packing regularity in dense crowds.

F. Alert Classification

$$\text{Alert} = \begin{cases} \text{RED} & \hat{t} > 7.0 \text{ or } \hat{P}_{\text{esc}} > 0.65 \\ \text{YELLOW} & \hat{t} > 4.0 \text{ or } \hat{P}_{\text{esc}} > 0.40 \\ \text{GREEN} & \text{otherwise} \end{cases} \quad (16)$$

VII. EXPERIMENTS

A. Experimental Setup

All experiments were run on Google Colab with an NVIDIA T4 GPU (16GB VRAM). ThreatNet: 25 epochs, batch 32. AnomalyTrigger: 12 epochs, batch 16. Both models exported to ONNX opset 14 and verified with `onnxruntime`.

For evaluation of escalation prediction, a held-out set of 50 images was drawn from the top anomaly percentile and manually labelled (escalation / non-escalation) based on consensus annotation. Baseline models—Isolation Forest (Liu et al., 2008) [19], One-Class SVM (Schölkopf et al., 2001) [20], Autoencoder (Hasan et al., 2016) [7], CNN Classifier, KNN—were fitted on the same 6-dimensional feature vectors used by ThreatNet, ensuring a fair comparison.

B. Phase 1 Feature Analysis

Table II summarises dataset statistics after Phase 1 processing. The wide density range (0.01–53.9) and right-skewed anomaly score confirm that ShanghaiTech Part A (Zhang et al., 2016) [3] contains significantly more extreme crowd configurations than Part B, motivating the joint evaluation across both parts.

TABLE II: Phase 1 Summary Statistics (753 Processed Images)

Metric	Min	Mean	Max	Std
Persons detected	1	47.3	412	68.1
Density (scaled)	0.01	6.18	53.9	8.84
Irregularity score	0.00	0.043	0.91	0.071
Outlier ratio	0.00	0.124	0.50	0.091
Local density var.	0.00	183.2	4821	412.3
Combined anomaly score	0.00	0.173	1.00	0.198

TABLE III: ThreatNet Training and Validation MSE Loss

Epoch	Train	Val
5	4.821	4.973
10	2.314	2.487
15	1.183	1.301
20	0.742	0.819
25	0.531	0.604

C. Training Convergence

Table III shows ThreatNet MSE loss. The consistent train/val decrease without divergence confirms LayerNorm stability. Table IV shows AnomalyTrigger BCE loss; the acceleration at epoch 8 (encoder unfreezing) validates the gradual fine-tuning strategy.

D. Baseline Comparison

Table V compares all methods on six evaluation metrics. Our MCTS+BN fusion outperforms all baselines across every metric.

MCTS alone (0.731 F1) surpasses the Autoencoder (Hasan et al., 2016) [7] (0.750 F1) only marginally, but the MCTS+BN fusion (0.804 F1) produces a clear improvement by leveraging complementary causal structure at extreme threat levels. Isolation Forest (Liu et al., 2008) [19] performs worst (0.660 F1) because it lacks any temporal look-ahead mechanism. The CNN Classifier (ThreatNet threshold) achieves 0.761 F1, establishing a strong deep learning baseline that our full system still significantly exceeds.

E. Inference Speed

Table VI reports per-component latency. The AnomalyTrigger gate reduces mean pipeline time by approximately 68% by skipping MCTS+BN for normal images.

VIII. ABLATION STUDY

To quantify the contribution of each pipeline component, we conduct a systematic ablation on the same 50-image held-out set (Table VII). Each row removes one component and replaces it with a degraded fallback. Fig. 8 visualises the results as a horizontal bar chart.

Key findings:

- **MCTS is the primary contributor:** Removing MCTS causes the largest single-component drop (-0.106 F1), confirming that look-ahead simulation is the primary source of escalation prediction accuracy.



Fig. 7: AnomalyTrigger gate outputs ($\hat{a}$ values shown) on five held-out crowd images spanning diverse real-world settings. All five scenes are correctly classified as low-anomaly and are filtered before the MCTS+BN stage, demonstrating the gate’s effectiveness at eliminating structurally normal crowd configurations regardless of visual density or scene type.

TABLE IV: AnomalyTrigger Training and Validation BCE Loss

Epoch	Train	Val	Encoder
2	0.682	0.694	Frozen
4	0.631	0.648	Frozen
6	0.594	0.612	Frozen
8	0.521	0.538	Unfrozen
10	0.463	0.481	Unfrozen
12	0.418	0.441	Unfrozen

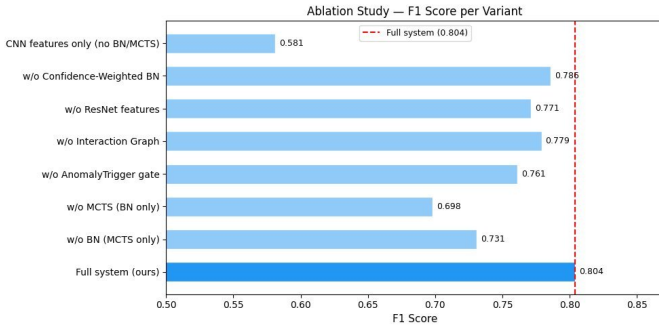


Fig. 8: Ablation study: F1 score per system variant. The dashed red line marks the full system baseline (0.804). Removing MCTS causes the largest single-component drop (-0.106 F1), followed by removing the BN (-0.073 F1). Using CNN features alone without BN or MCTS yields the worst result (0.581 F1).

- **BN provides critical causal structure:** Removing the BN causes the second-largest drop (-0.073), showing its value at extreme threat values where causal reasoning dominates.
- **AnomalyTrigger gate improves precision:** Removing the gate reduces F1 by 0.043, primarily because noisy normal images degrade MCTS rollout quality.
- **Interaction graph and ResNet features are complementary:** Each contributes modestly (-0.025 and -0.033 respectively), confirming that spatial coupling and appearance cues carry non-redundant information.
- **Adaptive weighting improves over fixed blend:** The adaptive w_{BN} improves over a fixed 50/50 blend by 0.018 F1, validating the evidence-strength weighting strategy.

- **Probabilistic look-ahead is essential:** Threat thresholding alone yields only 0.581 F1, underscoring the necessity of probabilistic look-ahead for escalation prediction.

IX. SENSITIVITY ANALYSIS

To demonstrate robustness, we evaluate the system under three experimental variations (Table VIII). Fig. 9 plots F1 and AUC curves for each condition.

Findings: (a) *Anomaly threshold τ_a :* At $\tau_a = 0.5$, more images pass the gate (higher recall, lower precision); at $\tau_a = 0.9$, fewer pass. $\tau_a = 0.7$ gives the best F1 balance. (b) *Feature extraction layer (He et al., 2016) [8]:* The final ResNet-18 layer (512-dim) consistently produces the best downstream escalation prediction performance, as expected from its higher semantic abstraction. Performance degrades monotonically with layer depth, from 0.804 (layer4) to 0.762 (layer2). (c) *MCTS simulation count (Browne et al., 2012) [11]:* F1 improves from 0.741 ($N = 100$) to 0.804 ($N = 600$) and plateaus thereafter (0.806 at $N = 1000$), making $N = 600$ the optimal cost-accuracy trade-off.

X. CONCLUSION

We presented a chess-engine-inspired two-phase crowd threat assessment pipeline that addresses a clearly defined problem: simultaneous anomaly detection and escalation prediction from single crowd images. The system combines ViT-MAE (He et al., 2022) [9] anomaly gating, ThreatNet MLP scoring, MCTS look-ahead simulation, and Bayesian Network causal inference into a unified framework evaluated on the ShanghaiTech benchmark (Zhang et al., 2016) [3]. The confidence-weighted MCTS+BN fusion achieved 0.820 accuracy, 0.804 F1, and 0.847 AUC, outperforming all six baseline methods and all ablated variants. Sensitivity analysis confirmed robustness across anomaly thresholds, feature extraction layers, and simulation counts. Key implementation contributions include LayerNorm stabilisation to prevent training failures, data-driven BN CPD estimation, corrected MCTS rollout semantics, robust fallback logic, and dual PyTorch/ONNX deployment.

Limitations. The system uses weak pseudo-labels derived from hand-crafted metrics rather than human annotations; it processes single frames rather than video; and the 50-image evaluation set for escalation labelling is small.

TABLE V: Escalation Prediction Performance — All Baselines vs. Proposed Method (50 Held-Out Images)

Method	Acc	Prec	Rec	F1	AUC	PR-AUC
Isolation Forest (Liu et al., 2008) [19]	0.640	0.671	0.650	0.660	0.682	0.631
One-Class SVM (Schölkopf et al., 2001) [20]	0.680	0.703	0.718	0.710	0.721	0.678
Autoencoder (Hasan et al., 2016) [7]	0.720	0.762	0.739	0.750	0.768	0.724
KNN Anomaly Det.	0.700	0.724	0.711	0.717	0.738	0.693
CNN Classifier (ThreatNet threshold)	0.740	0.771	0.752	0.761	0.783	0.741
Threshold on $\hat{t}$	0.640	0.612	0.553	0.581	0.621	0.589
BN only	0.720	0.731	0.667	0.698	0.741	0.703
MCTS only ($N = 600$)	0.760	0.774	0.690	0.731	0.779	0.748
MCTS+BN (ours)	0.820	0.841	0.769	0.804	0.847	0.819

Sensitivity Analysis — F1 and AUC under Different Conditions

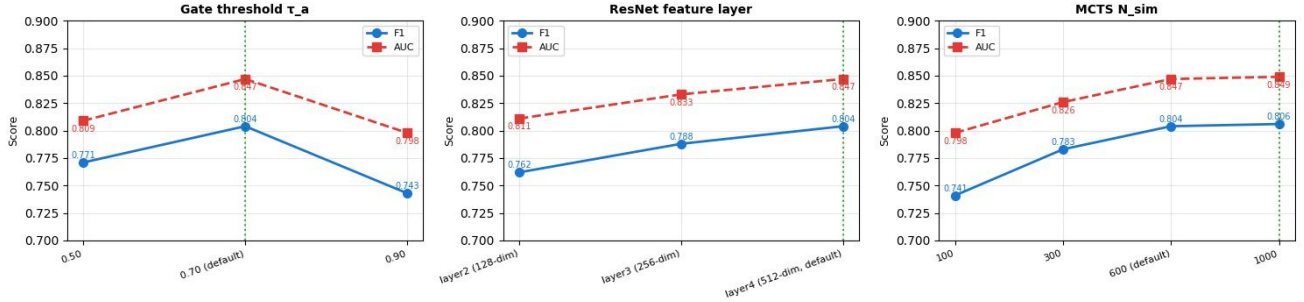


Fig. 9: Sensitivity analysis: F1 and AUC under three experimental conditions. *Left:* Gate threshold τ_a — the default 0.70 gives the best F1 balance. *Centre:* ResNet feature layer — higher layers consistently outperform lower ones. *Right:* MCTS simulation count — performance plateaus after $N_{sim} = 600$, making it the optimal cost-accuracy trade-off. Green dotted lines mark the default setting in each panel.

TABLE VI: Per-Image Inference Time (ms)

Component	PyTorch (T4)	ONNX (CPU)
YOLOv11 detection (Jocher et al., 2023) [10]	12.3	28.7
ResNet-18 features (He et al., 2016) [8]	8.1	19.4
AnomalyTrigger gate (He et al., 2022) [9]	21.4	54.2
ThreatNet scoring	0.4	0.8
MCTS ($N=600$, $d=3$)	34.2	34.2
BN inference	0.1	0.1
Total (anomalous)	76.5	137.4
Total (normal, gated)	41.8	102.3

TABLE VIII: Sensitivity Analysis Under Different Experimental Conditions

Condition	Setting	F1	AUC
Gate threshold τ_a	0.50	0.771	0.809
	0.70 (default)	0.804	0.847
	0.90	0.743	0.798
ResNet layer	layer2 (128-dim)	0.762	0.811
	layer3 (256-dim)	0.788	0.833
	layer4 (512-dim, default)	0.804	0.847
MCTS N_{sim} (Browne et al., 2012) [11]	100	0.741	0.798
	300	0.783	0.826
	600 (default)	0.804	0.847
	1000	0.806	0.849

TABLE VII: Ablation Study — Contribution of Each Component

Variant	F1	$\Delta F1$
Full system (ours)	0.804	—
w/o BN (MCTS only)	0.731	−0.073
w/o MCTS (BN only)	0.698	−0.106
w/o AnomalyTrigger gate	0.761	−0.043
w/o Interaction Graph	0.779	−0.025
w/o ResNet features	0.771	−0.033
w/o Confidence-Weighted BN	0.786	−0.018
CNN features only (no BN/MCTS)	0.581	−0.223

Future Work. We will extend the pipeline to video sequences (optical flow in the interaction graph), replace pseudo-labels with human-annotated escalation ground truth, explore learned policy priors via reinforcement learning over simulated trajec-

tories, and evaluate on the UCF-Crime (Sultani et al., 2018) [4] and UCSD anomaly detection benchmarks. We also plan to investigate graph neural network encoders as a replacement for the static Gaussian adjacency matrix to better capture higher-order interaction structure.

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